



**fit@hcmus**

TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN, ĐHQG - HCM  
KHOA CÔNG NGHỆ THÔNG TIN

# INTRODUCTION TO NATURAL LANGUAGE PROCESSING

## An Overview

TS. Nguyễn Hồng Bửu Long

- 1 Overview
- 2 Motivation & Applications
- 3 What is NLP?
- 4 Key Challenges in NLP
- 5 NLP Pipeline
- 6 Core NLP Tasks
- 7 Approaches to NLP

# 1. Overview

# Overview

## Content: 11 weeks

- Overview
- Linguistic Fundamentals for NLP
- Text Preprocessing & Representation
- Language Models
- Syntax & Parsing
- Text Classification
- Sequence-to-Sequence Models
- Advanced NLP with Transformers
- Applications & Future of NLP

## Assessments:

- Assignments (25%): Home exercises
- Projects (25%): Group project
- Exams (50%): Q&A on your group project
- Bonus marks: Q&A, Discussion...

## 2. Motivation & Applications

# Why NLP matters?

## Key points:

- Human communication = natural language
- Computers need to bridge the gap

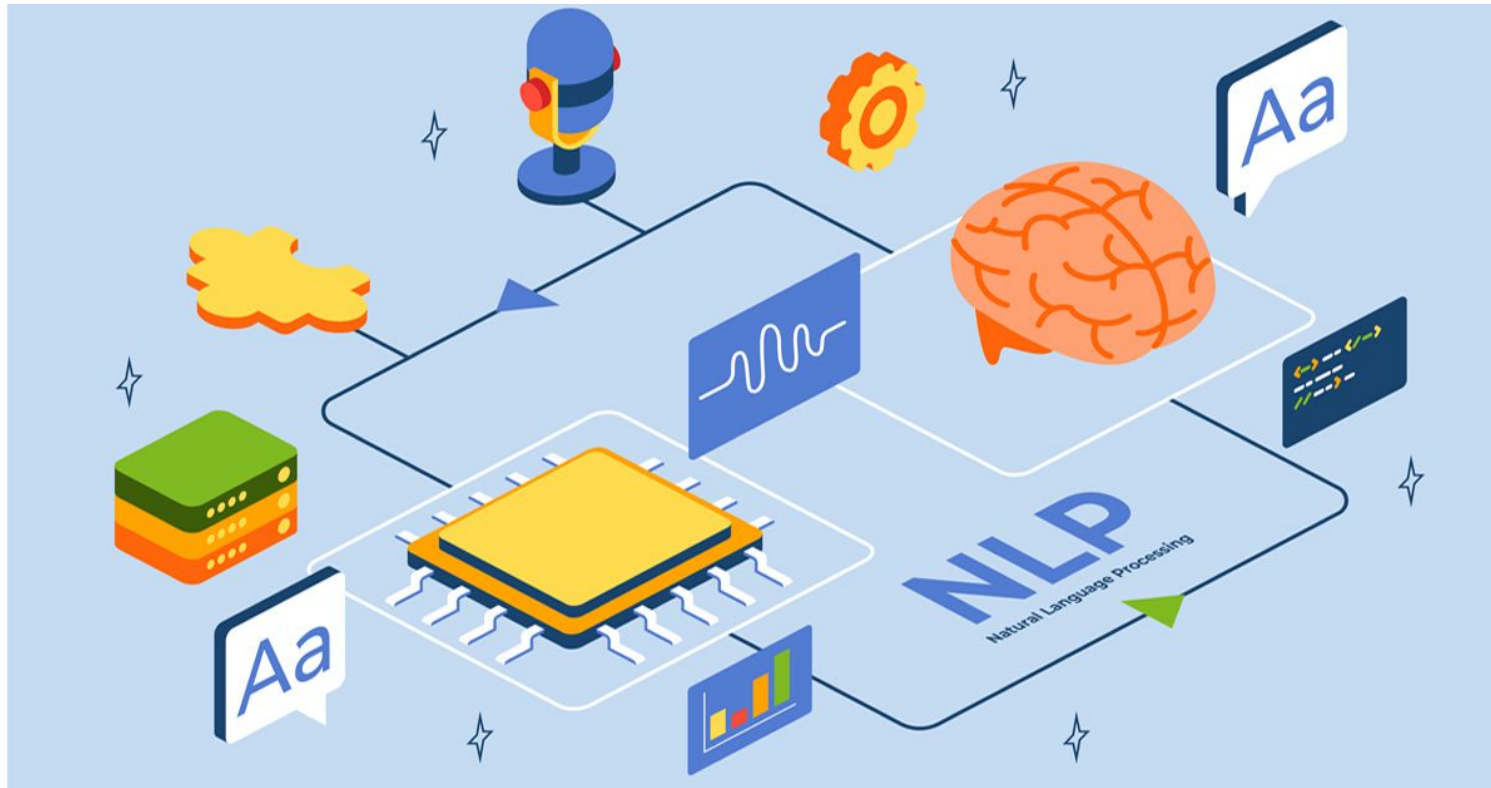
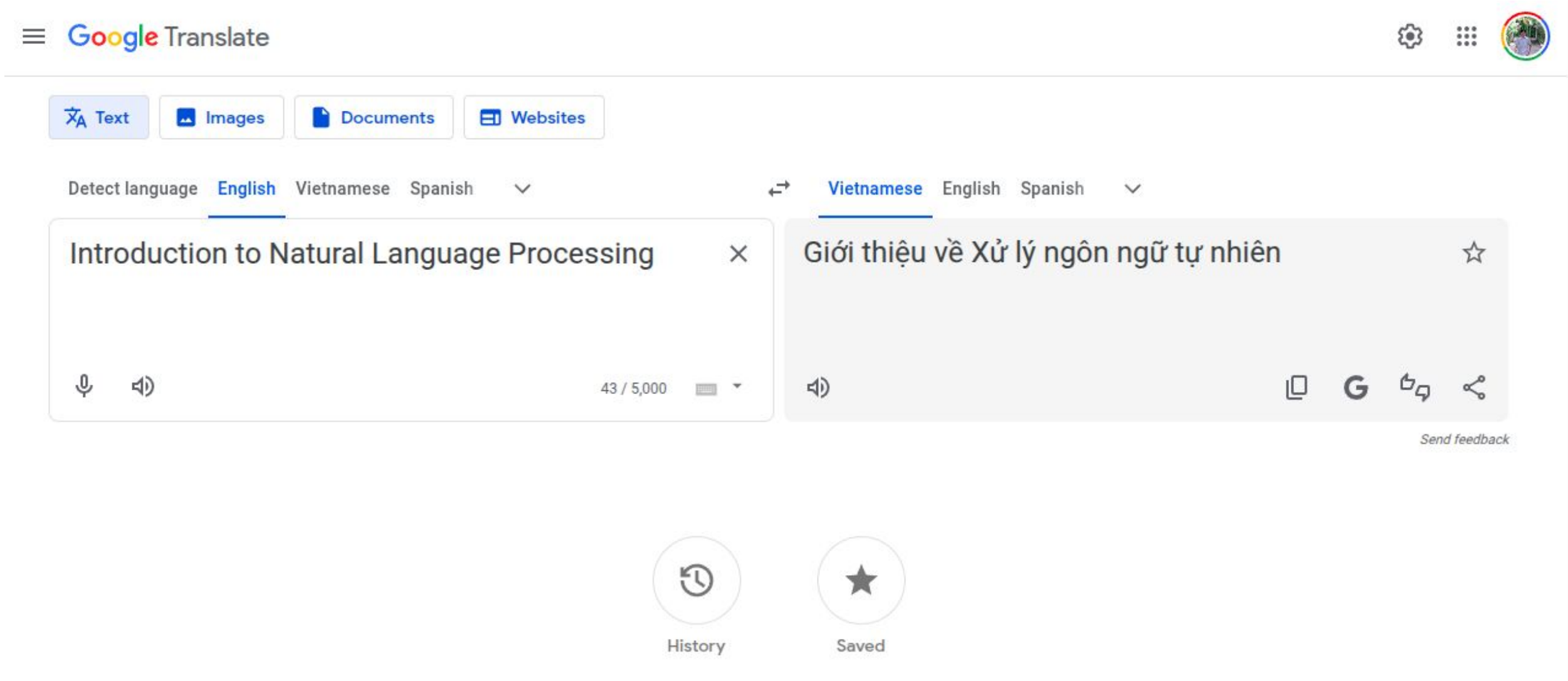


Image: <https://www.cwstechnology.com/blog/natural-language-processing/>

# Applications in daily life: Google Search



# Applications in daily life: Google Translate



# Applications in daily life: Google Assistant



Tổng quan

Tải Google Assistant

Tính năng sẵn có

Google Store

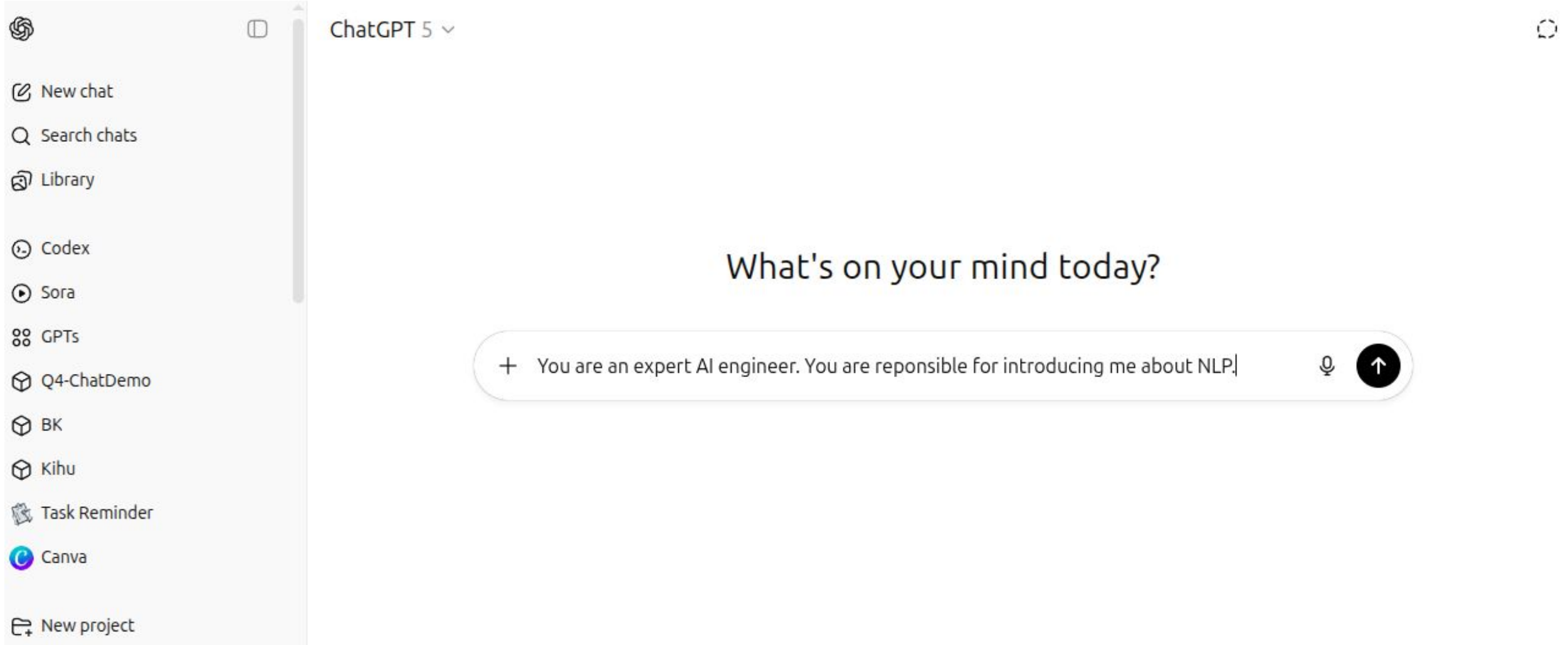


Xin chào, tôi có thể giúp gì cho bạn?

## Hãy gặp gỡ Google Assistant của bạn.

Đặt câu hỏi hay yêu cầu trợ giúp. Đây là trợ lý  
Google dành riêng cho bạn, luôn sẵn sàng hỗ trợ  
mỗi khi bạn cần.

# Applications in daily life: ChatGPT



# Applications in daily life: Gemini

☰

🔍

✎ New chat

LibreOffice Calc

Gems

📖 Storybook

🌟 Explore Gems

Recent

AI's Role in Scientific Resear...

Recipe Grocery Extraction

Deobfuscating JavaScript "...

Understanding AI Promptin...

AI for Studying and Learning

Qiskit Statevector Simulati...

⚙️ Settings & help

Gemini

2.5 Flash ▾

Upgrade

Hello, Nguyen Hong

Want to try out a few things?

Make my own custom mini figure

Turn me into a superhero

Give me an 80s style makeover

Create a professional headshot

You are an expert AI engineer. You are responsible for introducing me about NLP|

+ 🛠️ Tools

➤

# Applications in daily life: Kimhannom

Kim Sino Nom

Home

About Us

Typing instructions

Usage instructions

Privacy policy

Login



Kim Sino Nom

fit@hcmus



EN

Document category

Automatic

Sino/Nom

Automatic

REFRESH

DOCUMENT

IMAGE

Automatic

SINO - NOM



English

☐ Sino-Nom Typing

0 / 500

☐ Display image results

This work has been carried out by [the Faculty of Information Technology](#) and [the Computational Linguistics Center](#) at the [University of Science - VNUHCM](#) under the scientific research project of [the Department of Science and Technology in Ho Chi Minh City](#).

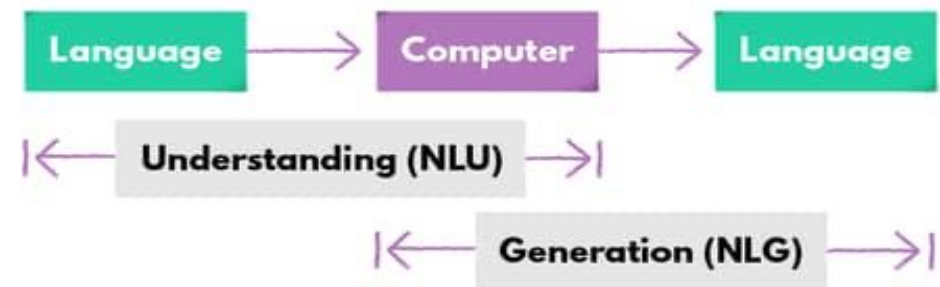
“Language is how humans interact. NLP enables machines to participate in this interaction.”

# 3. What is NLP?

# What is NLP?

## Key Points:

- Definition: **Computational methods** for understanding or generating natural language
- Goal: Allow computers to process, understand, and generate human language
- Components:
  - NLU – Understanding intent, meaning
  - NLG – Producing human-like text



# 4. Challenges in NLP

# Challenges in NLP

**Massive Challenge:** make the meaning (semantics, pragmatics) of human language accessible to computers.

- **Ambiguity:** same text can have very different meanings.
- **Different texts** can have **same meaning**, e.g. different wordings of summaries of the same document.
- **No hard rules:** socio-linguistic differences, colloquial and informal language (social media!).
- **So many languages!**
- **Code and language switching:** mixing standard language and dialect, mixing languages; often in emails and social media.

# Challenges in NLP

## Why it is difficult - Examples:

- "He went to the **bank**" (savings bank, river bank?)
- "If you love **money problems** show up" (love money or love money problems?);
- "He **wrote a report on Mars**" (report written being on Mars or report about Mars);
- "Cop kills man with knife" (who has the knife?)
- "Mark **saw the men with the telescope. He** had brown hair." (who had the telescope / brown hair?)
- Headline: "Teacher **Strikes Idle** Kids" (cred. Dan Klein)
- "This is **shit**" vs. "This is the **shit**"
- "@angry\_barista I baked you a cake but I ated it", "Why won't you show my location?! <http://twitpic.com/2y2es>",
- "@markhardy1974 Me too **#itm**" (ex. from kaggle Sentiment140 tweets dataset)
- "he wanted to do it but he got **cold feet** which made him **lose his face**" (multi-word expressions)
- "**It is cold**" (statement or request for action → pragmatics)

⇒ Often clear from context or with background knowledge but system has to deal with it

# Challenges in NLP

## non-standard English

Great job @justinbieber! Were SOO PROUD of what youve accomplished! U taught us 2 #neversaynever & you yourself should never give up either♥

## segmentation issues

the New York-New Haven Railroad  
the New York-New Haven Railroad

## idioms

dark horse  
get cold feet  
lose face  
throw in the towel

## neologisms

unfriend  
Retweet  
bromance

## world knowledge

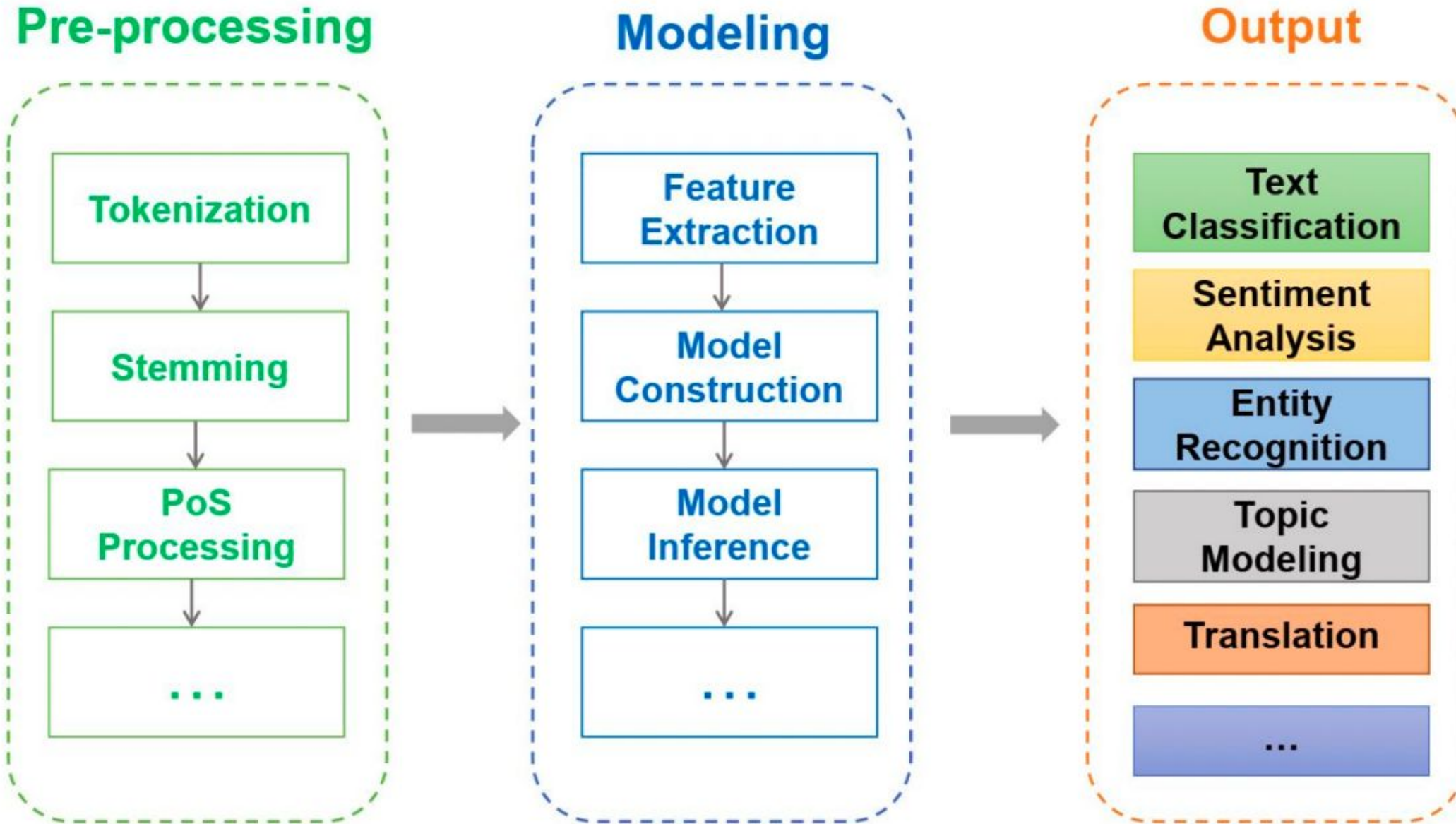
Mary and Sue are sisters.  
Mary and Sue are mothers.

## tricky entity names

Where is *A Bug's Life* playing ...  
*Let It Be* was recorded ...  
... a mutation on the *for* gene ...

# 5. NLP Pipeline

# NLP Pipeline: Typical pipeline

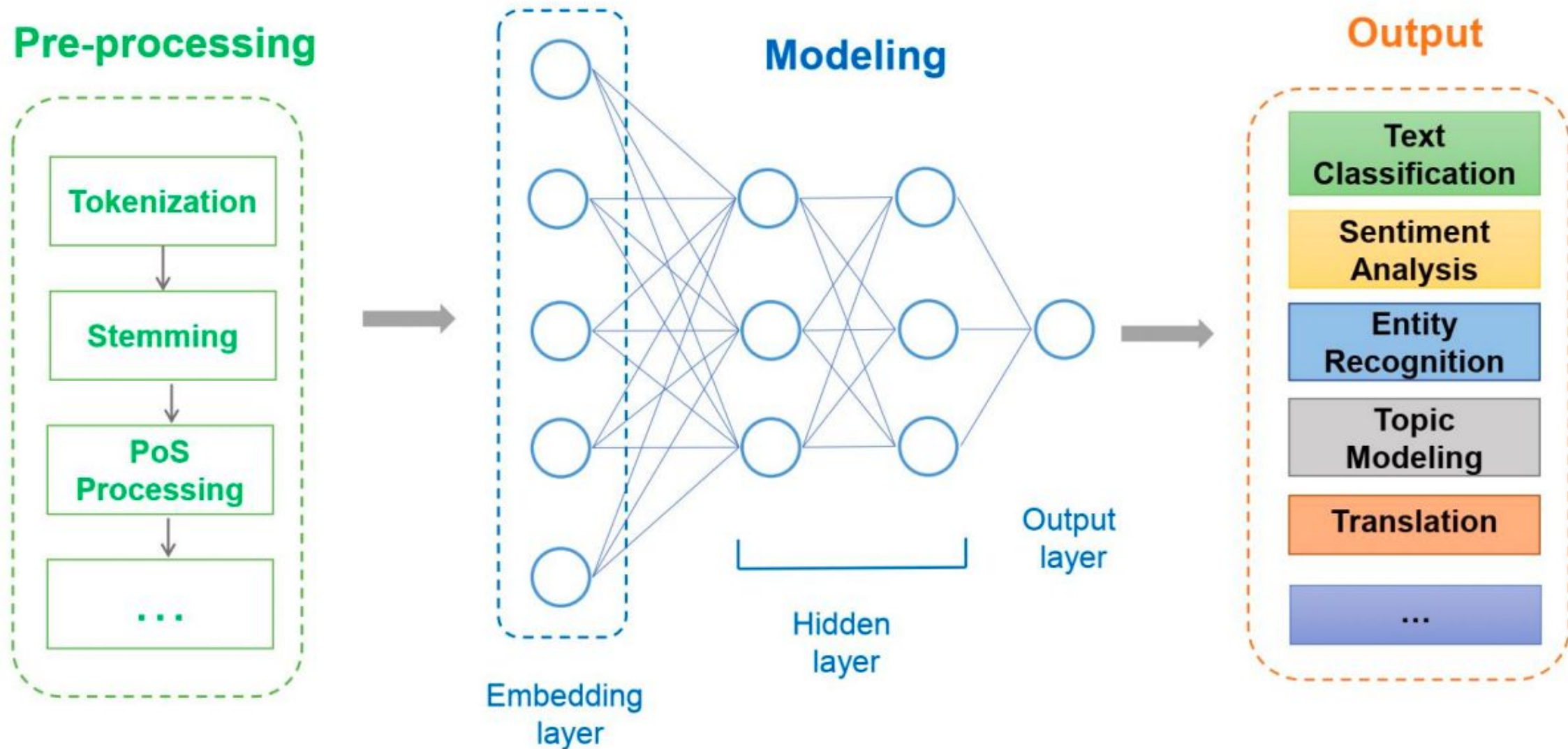


# NLP Pipeline: Pre-processing

## Key points:

- Text Input
- Preprocessing
  - Tokenization
  - Normalization (stemming, lemmatization)
  - Stop-word removal
  - POS tagging
- Feature Representation
  - Bag of Words / TF-IDF

# NLP Pipeline: DL-based pipeline

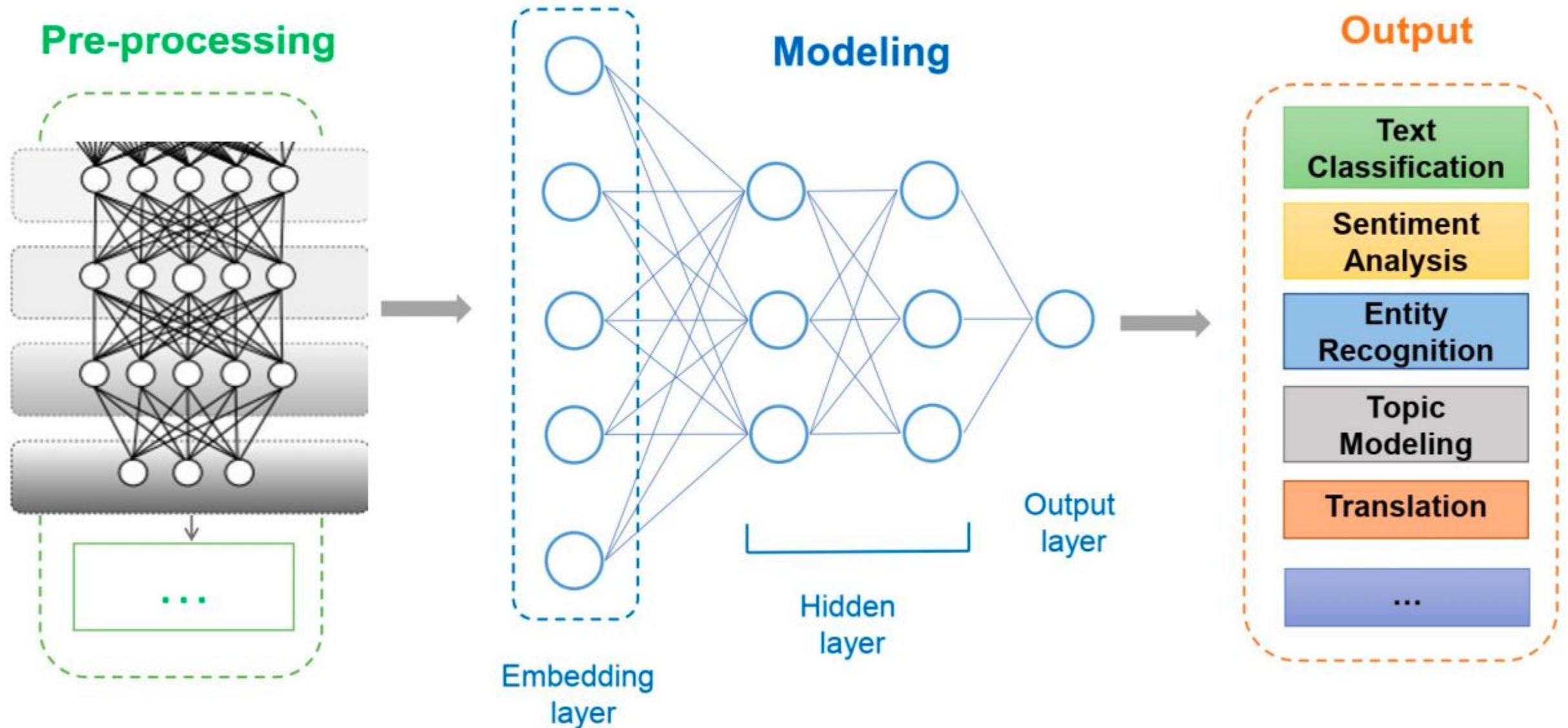


# NLP Pipeline: Pre-processing

## Key points:

- Text Input
- Preprocessing
  - Tokenization
  - Normalization (stemming, lemmatization)
  - Stop-word removal
  - POS tagging
- Feature Representation
  - Bag of Words / TF-IDF
  - Word embeddings
  - Contextual embeddings

# NLP Pipeline: LM-based pipeline



# 6. Core NLP Tasks

# Core NLP Tasks

Dan Jurafsky



## Language Technology

making good progress

mostly solved

### Spam detection

Let's go to Agra! ✓

Buy V1AGRA ... ✗

### Part-of-speech (POS) tagging

ADJ ADJ NOUN VERB ADV

Colorless green ideas sleep furiously.

### Named entity recognition (NER)

PERSON ORG LOC

Einstein met with UN officials in Princeton

### Sentiment analysis

Best roast chicken in San Francisco! 👍

The waiter ignored us for 20 minutes. 👎

### Coreference resolution

Carter told Mubarak he shouldn't run again.

### Word sense disambiguation (WSD)

I need new batteries for my *mouse*.

### Parsing

I can see Alcatraz from the window!

### Machine translation (MT)

第13届上海国际电影节开幕...

The 13<sup>th</sup> Shanghai International Film Festival...

### Information extraction (IE)

You're invited to our dinner party, Friday May 27 at 8:30



still really hard

### Question answering (QA)

Q. How effective is ibuprofen in reducing fever in patients with acute febrile illness?

### Paraphrase

XYZ acquired ABC yesterday

ABC has been taken over by XYZ

### Summarization

The Dow Jones is up

The S&P500 jumped

Housing prices rose

Economy is good

### Dialog

Where is Citizen Kane playing in SF?

Castro Theatre at 7:30. Do you want a ticket?



# Spam detection

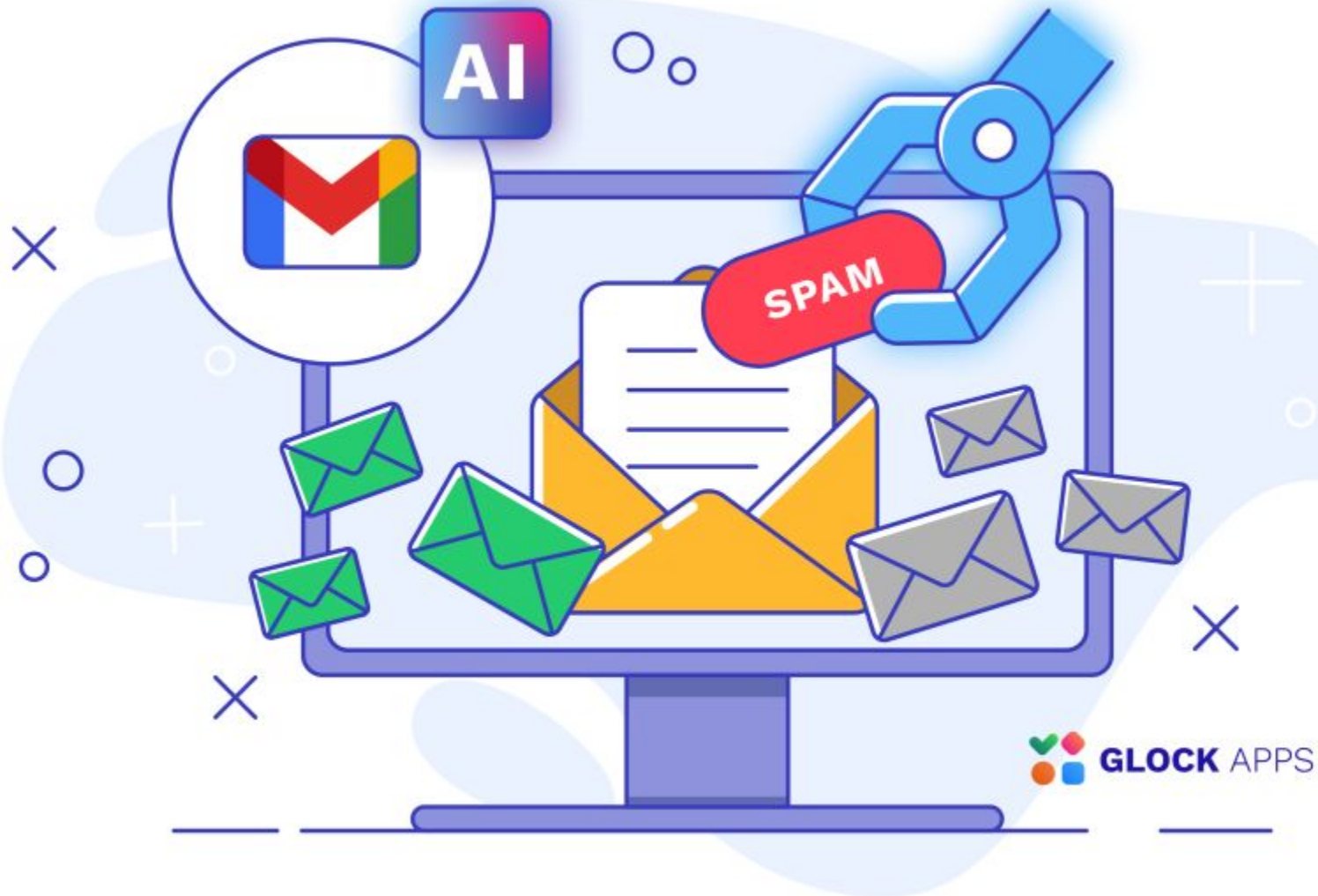


Image: <https://glockapps.com/blog/gmails-biggest-security-update-ai-powered-spam-filter/>

# Part-of-speech (POS) tagging

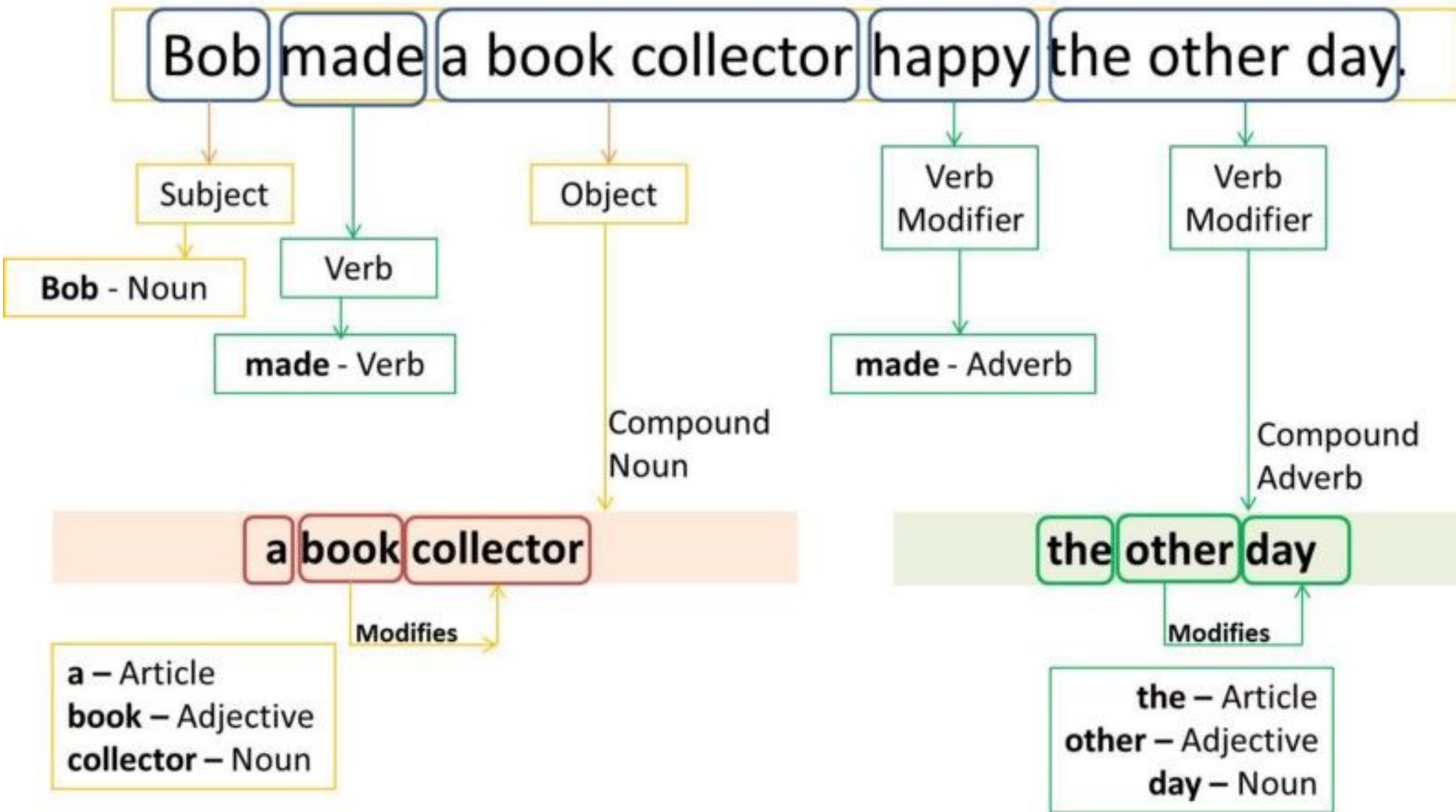


Image: <https://www.freecodecamp.org/news/an-introduction-to-part-of-speech-tagging-and-the-hidden-markov-model-953d45338f24/>

# Named Entity Recognition (NER)

In fact, the **Chinese** **NORP** market has the **three** **CARDINAL** most influential names of the retail and tech space – **Alibaba** **GPE** , **Baidu** **ORG** , and **Tencent** **PERSON** (collectively touted as **BAT** **ORG** ), and is betting big in the global **AI** **GPE** in retail industry space . The **three** **CARDINAL** giants which are claimed to have a cut-throat competition with the **U.S.** **GPE** (in terms of resources and capital) are positioning themselves to become the ‘future **AI** **PERSON** platforms’. The trio is also expanding in other **Asian** **NORP** countries and investing heavily in the **U.S.** **GPE** based **AI** **GPE** startups to leverage the power of **AI** **GPE** . Backed by such powerful initiatives and presence of these conglomerates, the market in APAC AI is forecast to be the fastest-growing **one** **CARDINAL** , with an anticipated **CAGR** **PERSON** of **45%** **PERCENT** over **2018 - 2024** **DATE** .

To further elaborate on the geographical trends, **North America** **LOC** has procured **more than 50%** **PERCENT** of the global share in **2017** **DATE** and has been leading the regional landscape of **AI** **GPE** in the retail market. The **U.S.** **GPE** has a significant credit in the regional trends with **over 65%** **PERCENT** of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as **Google** **ORG** , **IBM** **ORG** , and **Microsoft** **ORG** .

Image: <https://devopedia.org/named-entity-recognition>

# Sentiment Analysis



## Review



# Coreference Resolution

mention

**John** told **Sally** that **she** should come watch **him** play the **violin**.

antecedent

John told Sally that **she** should come watch him play the violin.  


coreferent

John told Sally that **she** should come watch him play the violin.  


cluster

**John** told **Sally** that **she** should come watch **him** play the **violin**.

anaphoric

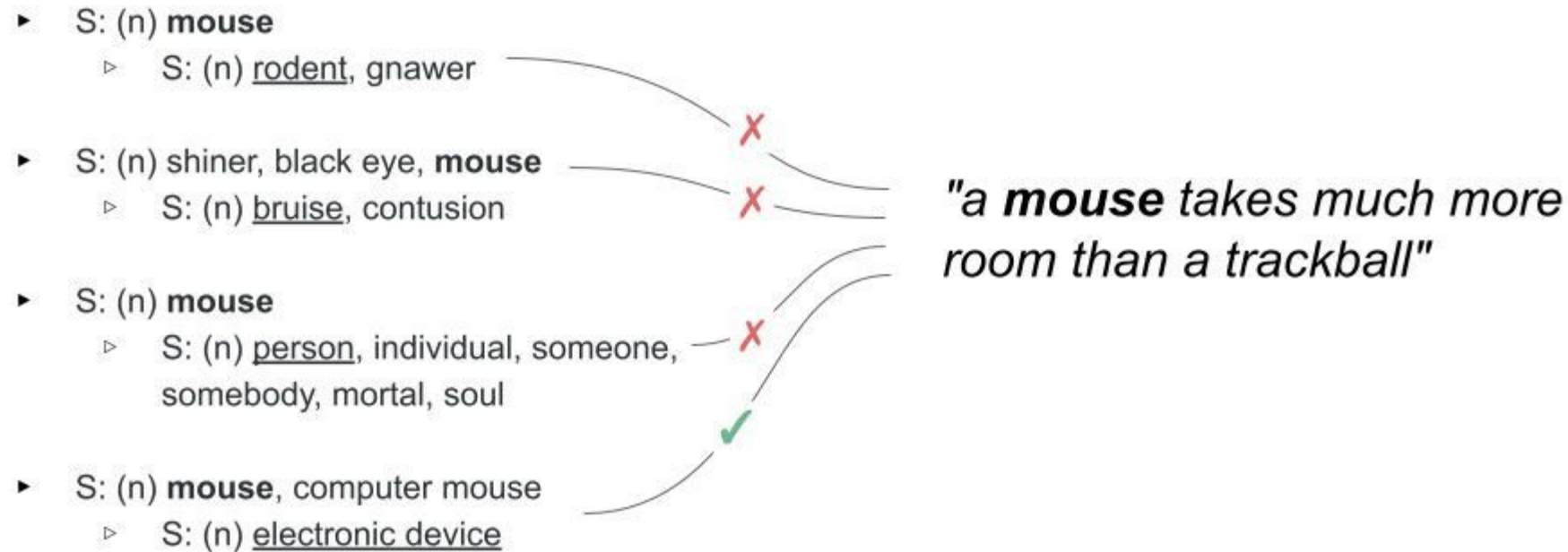
John told Sally that she should come watch him play the violin.  


non-anaphoric

John told Sally that she should come watch him play the **violin**.  


Image: <https://lumosbusiness.com/how-valuable-is-sentiment-analysis-for-brands/>

# Word Sense Disambiguation (WSD)



# Parsing

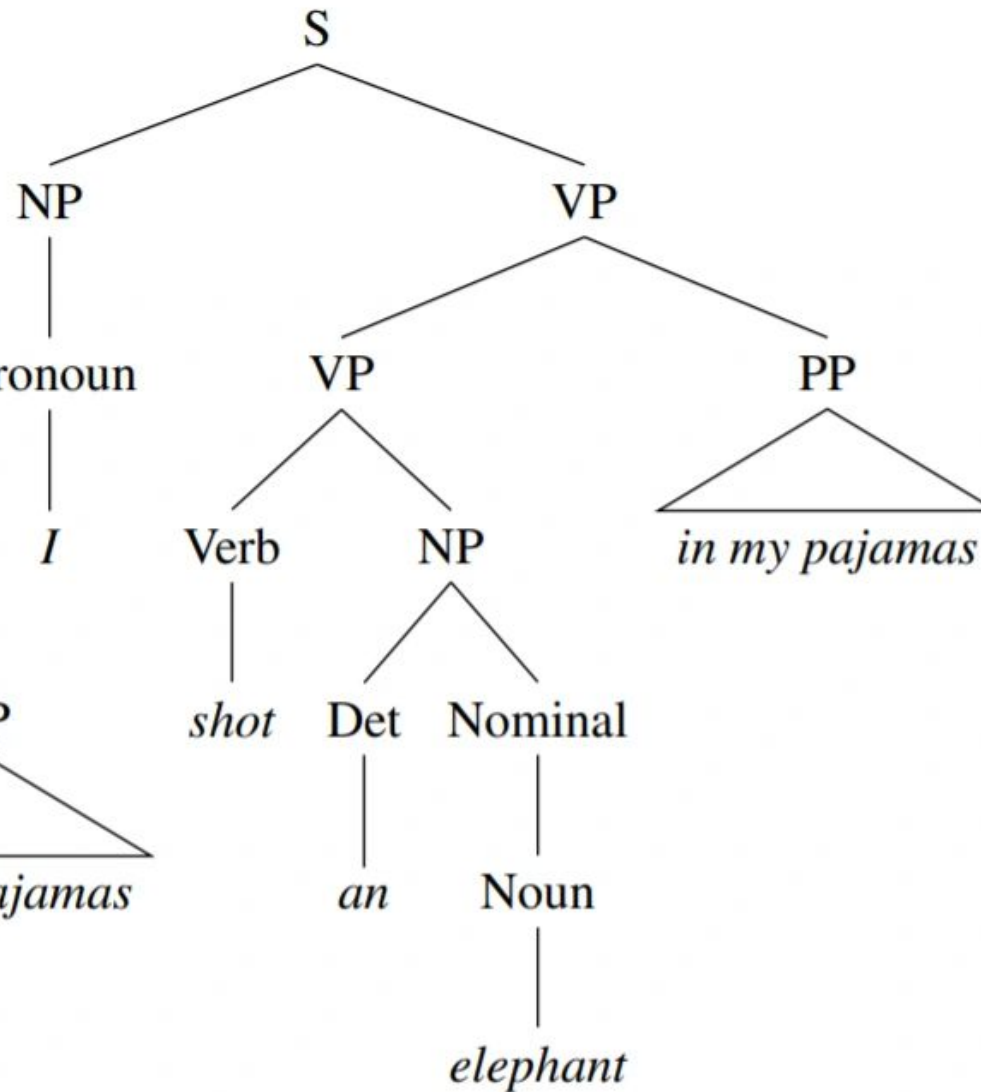
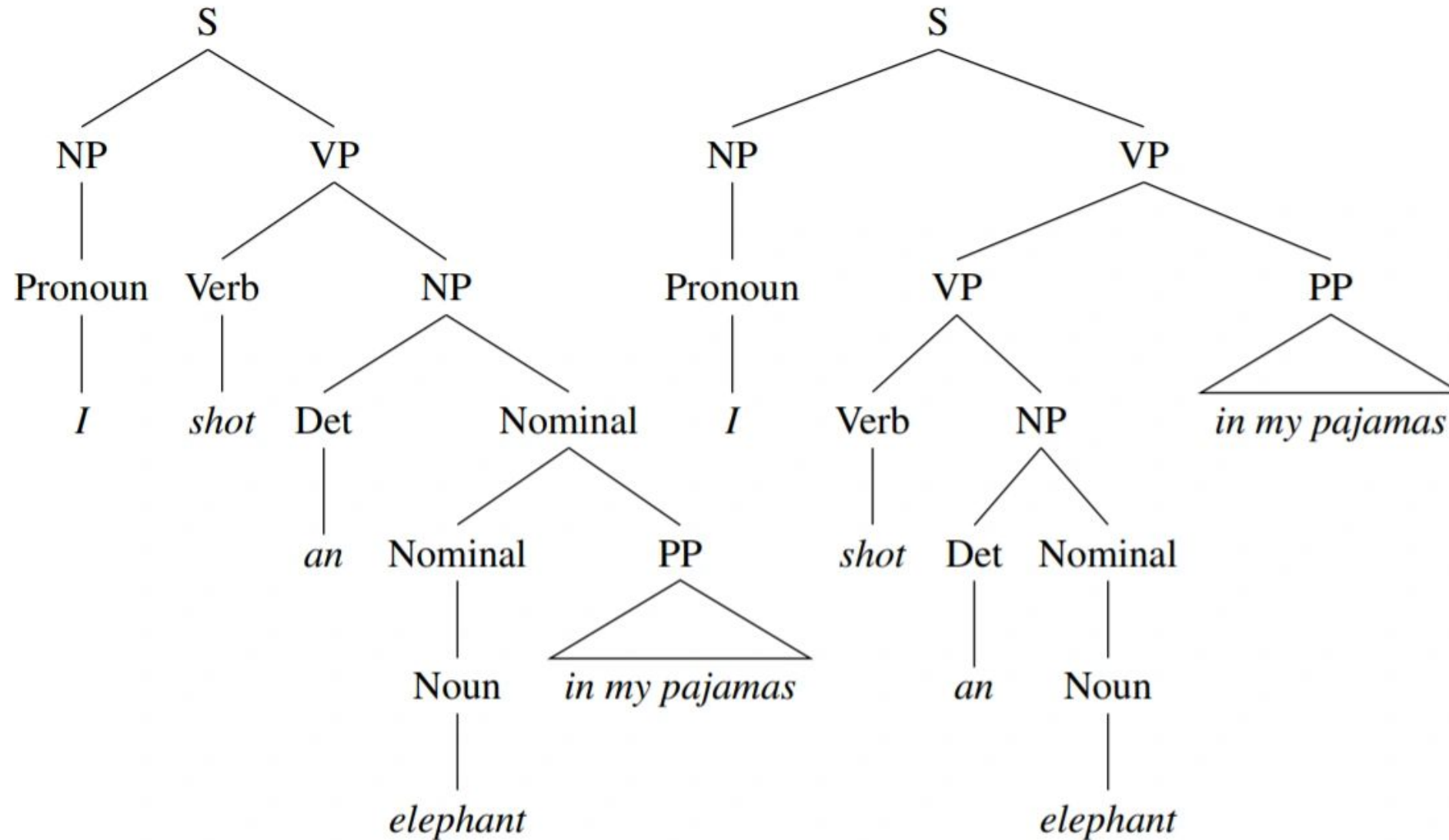


Image: <https://www.analyticsvidhya.com/blog/2021/12/dependency-parsing-in-natural-language-processing-with-examples/>

# Machine Translation (MT)

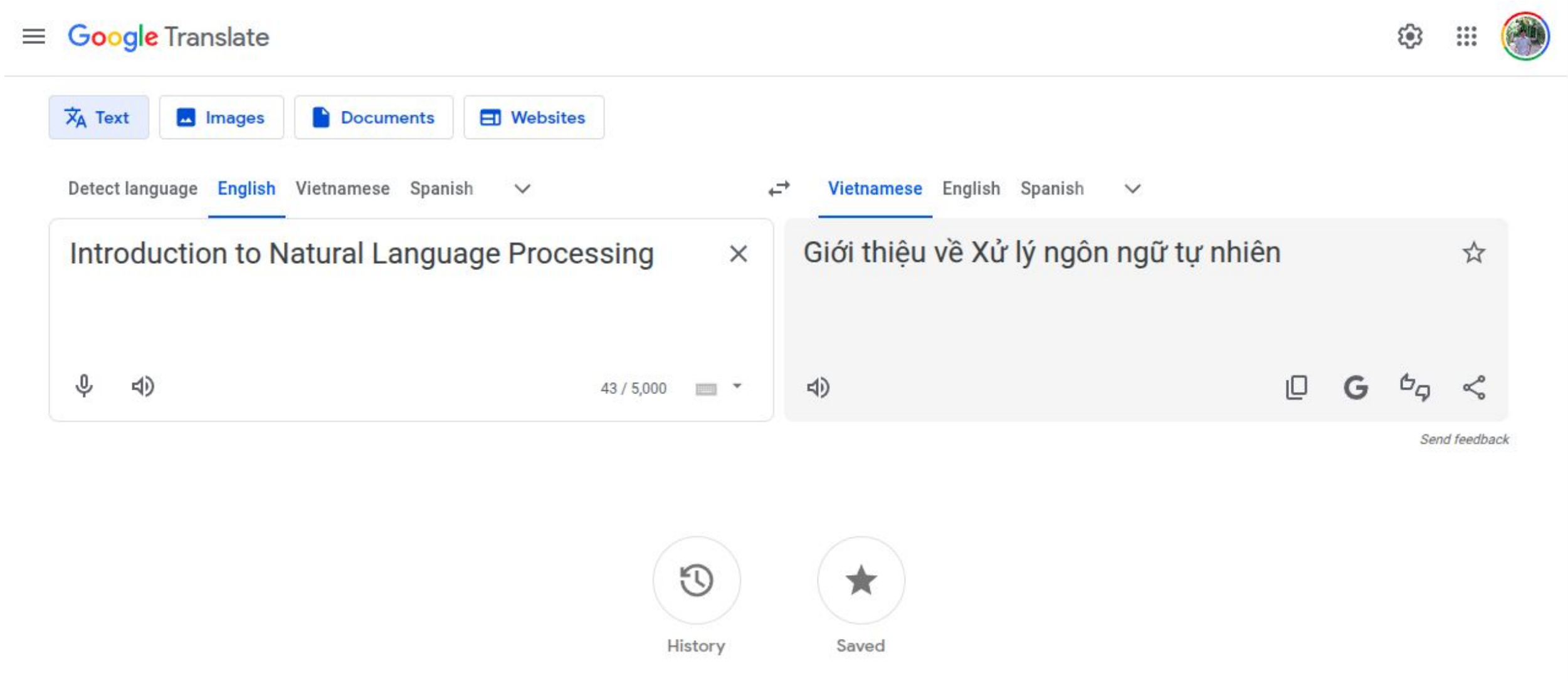
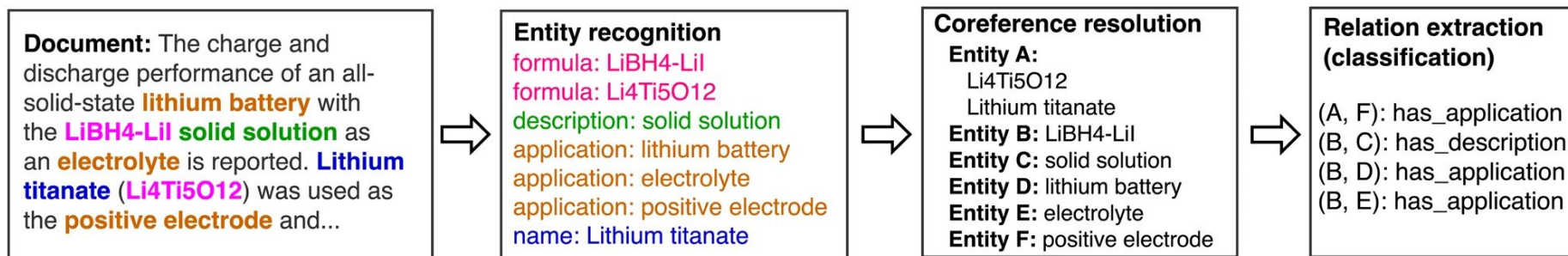


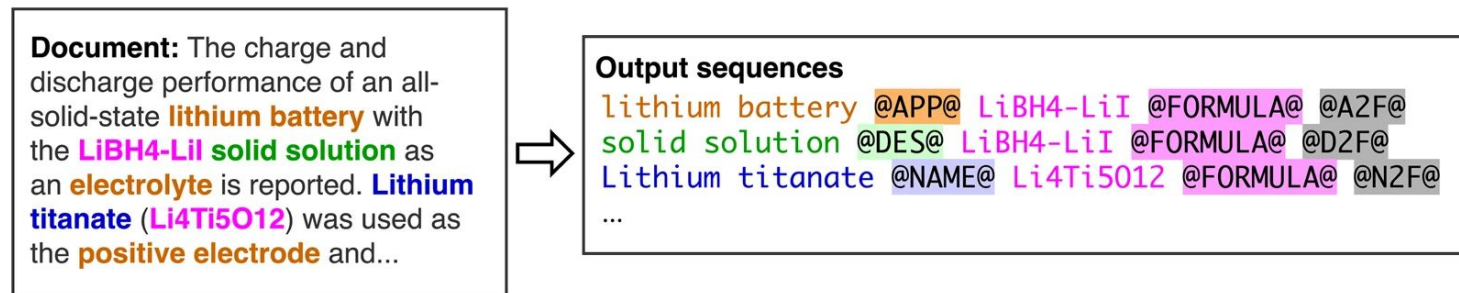
Image: <https://www.analyticsvidhya.com/blog/2021/12/dependency-parsing-in-natural-language-processing-with-examples/>

# Information Extraction (IE)

(a) **Previous:** Multi-step (pipeline) named entity recognition and relationship extraction



(b) **Previous:** seq2seq enumerates relationships as 2-tuples



(c) **This work:** Hierarchical entity relationships without explicit enumeration

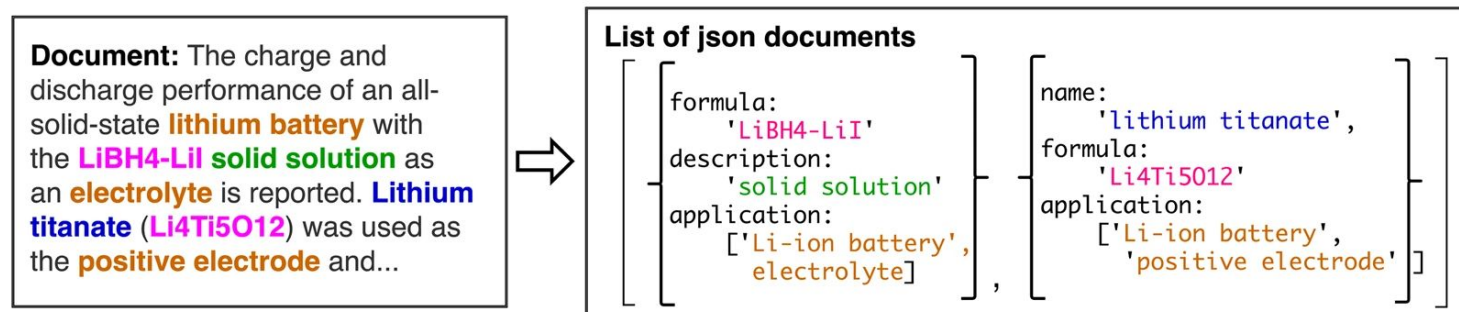


Image: <https://www.nature.com/articles/s41467-024-45563-x/figures/1>

# Question Answering (QA)

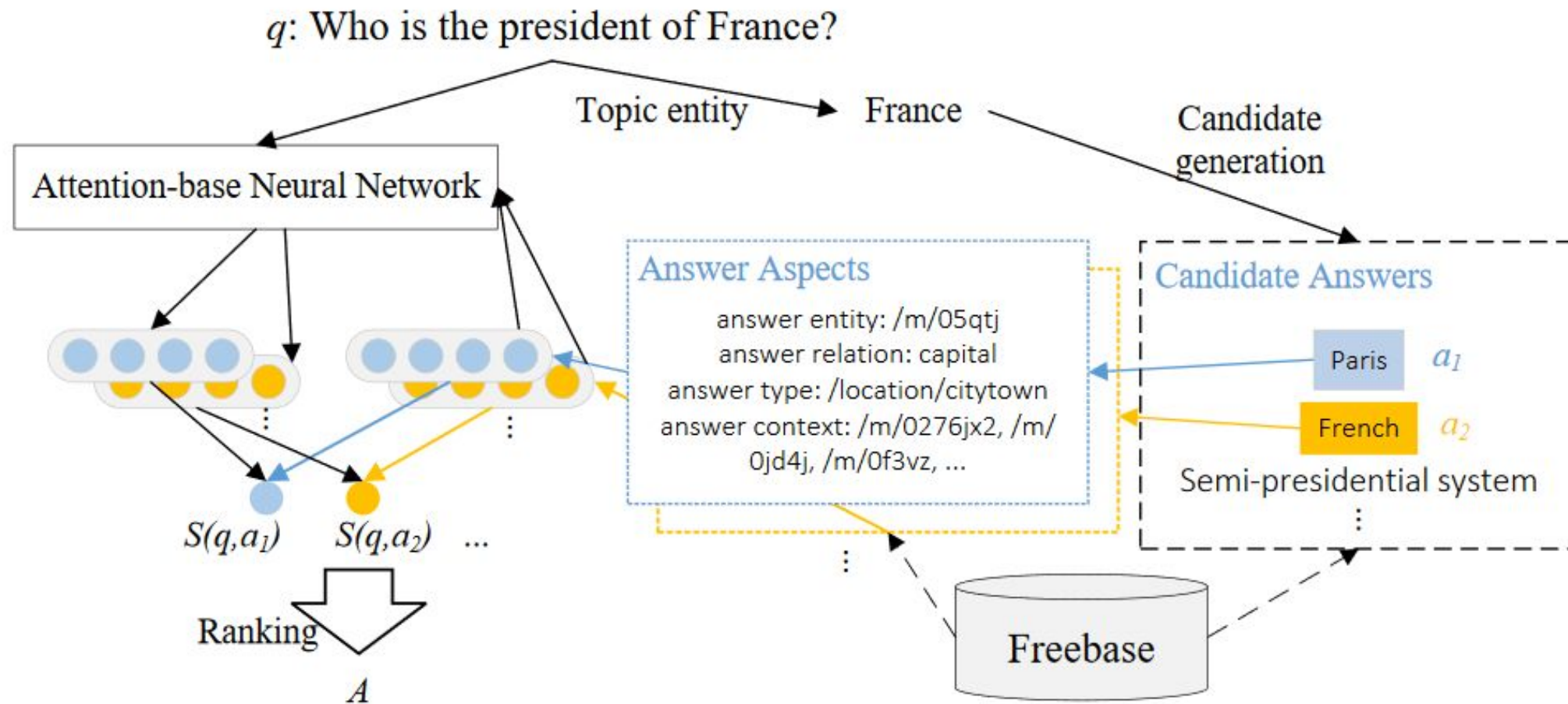


Image: <https://devopedia.org/question-answering>

# Paraphrase

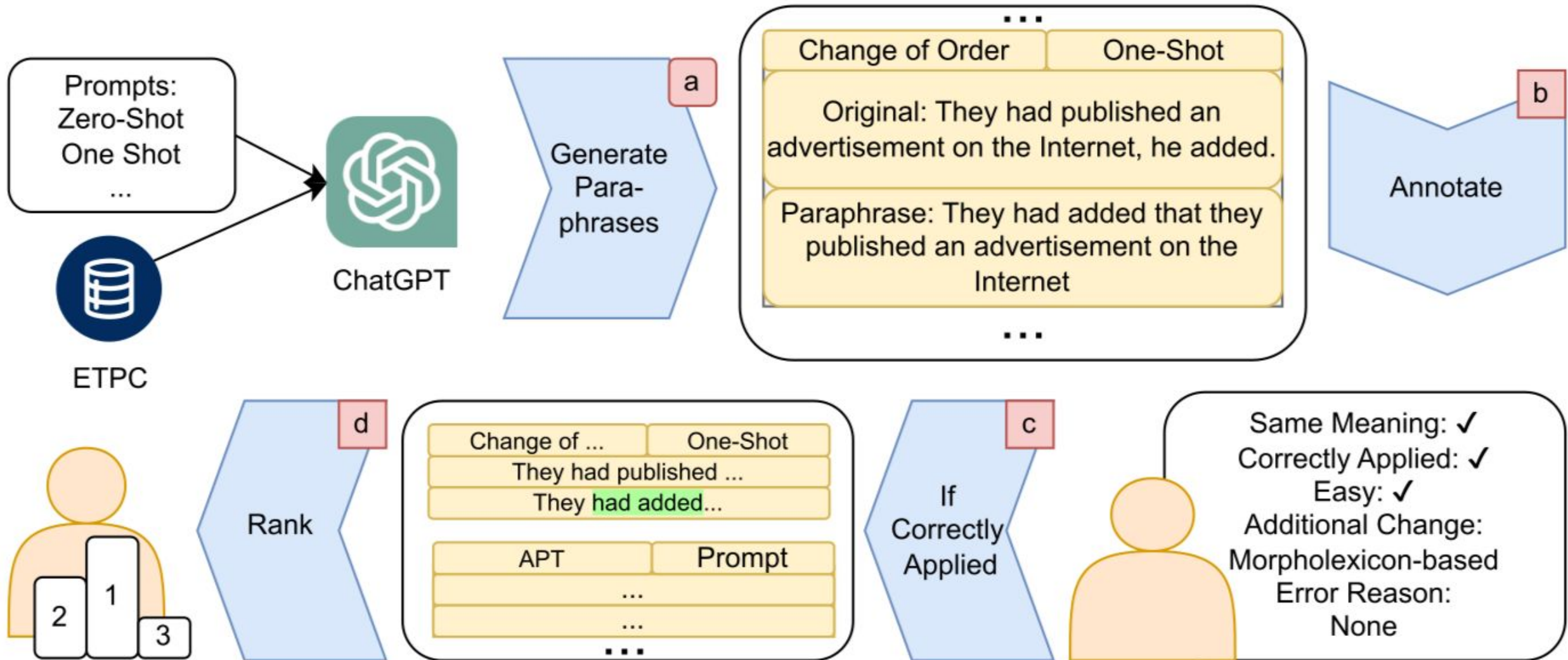
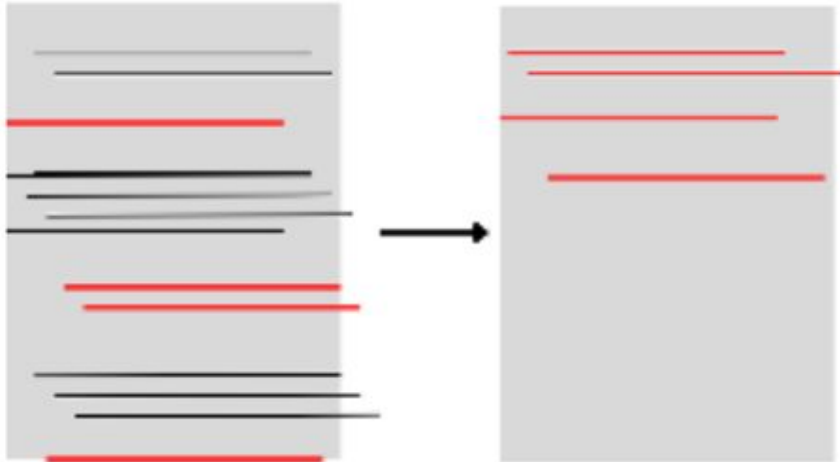


Image: <https://devopedia.org/question-answering>

# Summarization

## NLP Text Summarization

### Extractive



### Abstractive

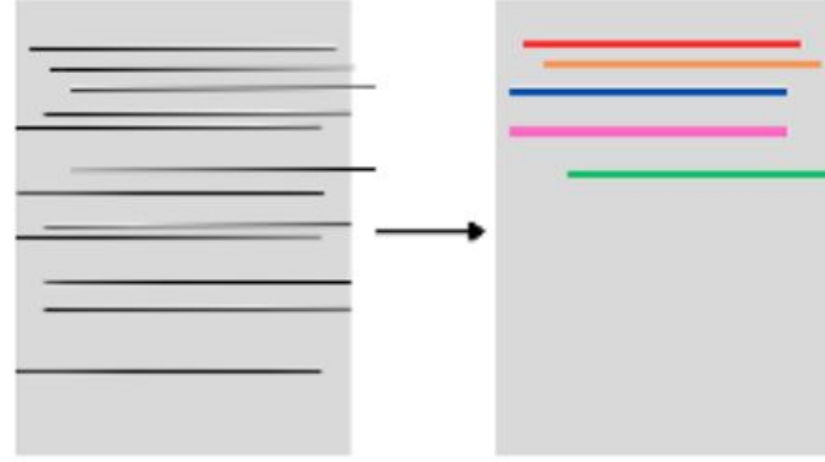


Image: <https://devopedia.org/question-answering>

# Dialog/Chatbot/Conversation

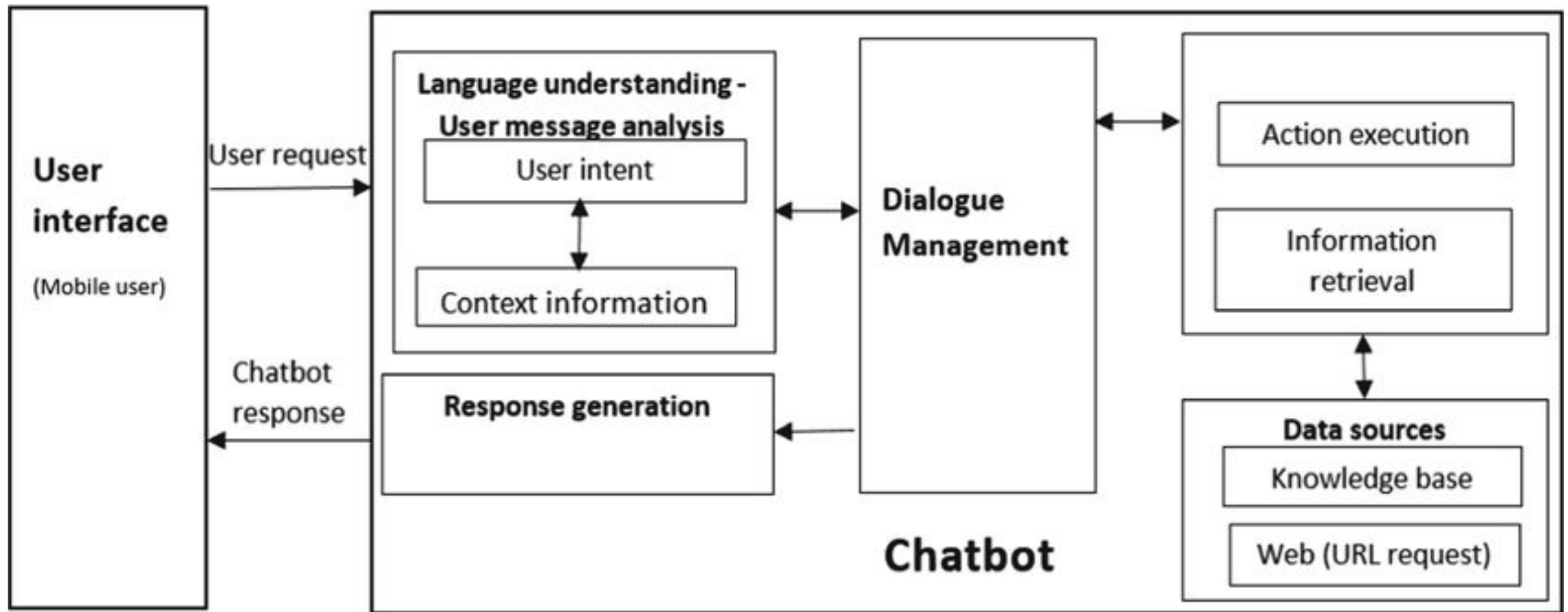


Image: <https://devopedia.org/question-answering>

# 7. Approaches to NLP

# Approaches to NLP

## Approaches:

- **Rule-based approaches:** Expert-written grammar rules
- **Statistical approaches:** Probabilistic models (n-grams, HMMs)
- **Deep Learning approaches:**
  - RNNs, LSTMs, GRUs
  - Attention & Transformers
  - Pretrained models (BERT, GPT, T5)
  - Large Pretrained models/Large Language models
  - World models

# 8. References

# References

- Nhập môn Xử lý ngôn ngữ tự nhiên, Đinh Điền, 2006
- Speech and Language Processing: <https://web.stanford.edu/~jurafsky/slp3/ed3book.pdf>
- Natural Language Toolkit: <https://www.nltk.org/>